Unit 1 AP Topics 14

1. Let f be the function given by $f(x) = \frac{\ln x}{x}$ for all $x > 0$. The derivative of f is given by $f'(x) = \frac{1-\ln x}{x^2}$.

(a) Write an equation for the line tangent to the graph of f at $x = e^2$.

(b) Find the x-coordinate of the critical point of f. Determine whether this point is a relative minimum, a relative maximum, or neither for the function f. Justify your answer.

(c) The graph of the function f has exactly one point of inflection. Find the x-coordinate of this point.

(d) Find $\lim\limits_{x \to 0^+} f(x)$.

Part A

The response can earn up to 2 points:

1 point is earned for $f(e^2)$ and $f'(e^2)$

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2},\ f'(e^2) = \frac{1-\ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

1 point is earned for correct answer

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$.

0	1	2

The response earns all of the following points:

1 point is earned for $f(e^2)$ and $f'(e^2)$

$$f(e^2) = \frac{\ln e^2}{e^2} = \frac{2}{e^2},\ f'(e^2) = \frac{1-\ln e^2}{(e^2)^2} = -\frac{1}{e^4}$$

1 point is earned for correct answer

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}(x - e^2)$.

Part B

The response can earn up to 3 points:

1 point is earned for $x = e$

1 point is earned for the relative maximum

1 point is earned for the justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.

0	1	2	3

Unit 1 AP Topics 14

The response earns all of the following points:

1 point is earned for $x = e$

1 point is earned for the relative maximum

1 point is earned for the justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.

Part C

The response can earn up to 3 points:

2 points are earned for $f''(x)$

$$f''(x) = \frac{-\frac{1}{x}x^2-(1-\ln x)2x}{x^4} = \frac{-3+2\ln x}{x^3} \text{ for all } x > 0$$
$$f''(x) = 0 \text{ when } -3 + 2\ln x = 0$$
$$x = e^{3/2}$$

1 point is earned for the correct answer

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

0	1	2	3

The response earns all of the following points:

2 points are earned for $f''(x)$

$$f''(x) = \frac{-\frac{1}{x}x^2-(1-\ln x)2x}{x^4} = \frac{-3+2\ln x}{x^3} \text{ for all } x > 0$$
$$f''(x) = 0 \text{ when } -3 + 2\ln x = 0$$
$$x = e^{3/2}$$

1 point is earned for the correct answer

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

Part D

The response can earn up to 1 point:

1 point is earned for the correct answer

$$\lim_{x\to 0^+} \frac{\ln x}{x} = -\infty \text{ or Does Not Exist}$$

Unit 1 AP Topics 14

0	1

The response earns all of the following point:

1 point is earned for the correct answer

$\lim_{x \to 0^+} \frac{\ln x}{x} = -\infty$ or Does Not Exist

2. Let f be the function given by the $f(x) = \frac{\ln x}{x}$ for all $x > 0$. The derivative of f is given by $f'(x) = \frac{1 - \ln x}{x^2}$.

Find $\lim_{x \to 0^+} f(x)$.

Part D

The response can earn up to 1 point:

1 point: for the correct answer

$\lim_{x \to 0^+} \frac{\ln x}{x} = -\infty$ or Does Not Exist

0	1

The response earns the following point:

1 point: for the correct answer

$\lim_{x \to 0^+} \frac{\ln x}{x} = -\infty$ or Does Not Exist

Unit 1 AP Topics 14

3. Let f be the function given by $f(x) = \ln|x/1 + x^2|$.

(a) Find the domain of f.

(b) Determine whether f is an even function, an odd function, or neither. Justify your conclusion.

(c) At what values of x does f have a relative maximum or a relative minimum? For each such x, use the first derivative test to determine whether $f(x)$ is a relative maximum or a relative minimum.

(d) Find the range of f.

Part A

1 point is earned for answer

$x \neq 0$

0	1

The student response earns all of the following points:

1 point is earned for answer

$x \neq 0$

Part B

1 point is earned for the answer

1 point is earned for identification using $f(x)$

Even

$$f(-x) = \ln\left|\frac{-x}{1+(-x)^2}\right| = f(x)$$

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for identification using $f(x)$

Even

$$f(-x) = \ln\left|\frac{-x}{1+(-x)^2}\right| = f(x)$$

Unit 1 AP Topics 14

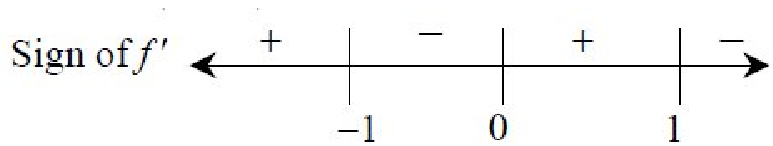
Part C

2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of $f'(x)$ and conclusion

$$f'(x) = \frac{1+x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$
$$= \frac{1-x^2}{x(1+x^2)}$$



$f(x)$ has relative max at $x = 1$

$f(x)$ has relative max at $x = -1$

0	1	2	3	4

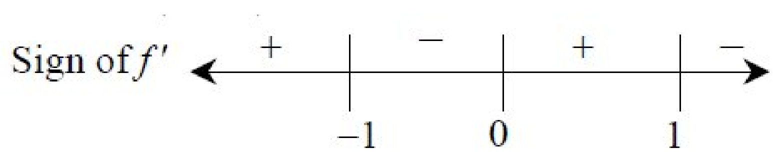
The student response earns all of the following points:

2 points are earned for the derivative

1 point is earned for finding critical number

1 point is earned for sign analysis of $f'(x)$ and conclusion

$$f'(x) = \frac{1+x^2}{x} \cdot \frac{(1+x^2)-2x^2}{(1+x^2)^2}$$
$$= \frac{1-x^2}{x(1+x^2)}$$



$f(x)$ has relative max at $x = 1$

$f(x)$ has relative max at $x = -1$

Part D

2 points are earned for the answer

1 point **deducted** for error

Unit 1 AP Topics 14

max is

$$f(1) = \ln \frac{1}{2} = -\ln 2$$
$$\lim_{x \to 0^+} f(x) = -\infty$$
$$\left(\text{or} \lim_{x \to 0^-} f(x) = \lim_{x \to \infty} f(x) = \lim_{x \to -\infty} f(x) = -\infty\right)$$
$$\therefore \text{range} = \left(-\infty, \ln \frac{1}{2}\right]$$

0	1	2

The student response earns all of the following points:

2 points are earned for the answer

1 point **deducted** for error

max is

$$f(1) = \ln \frac{1}{2} = -\ln 2$$
$$\lim_{x \to 0^+} f(x) = -\infty$$
$$\left(\text{or} \lim_{x \to 0^-} f(x) = \lim_{x \to \infty} f(x) = \lim_{x \to -\infty} f(x) = -\infty\right)$$
$$\therefore \text{range} = \left(-\infty, \ln \frac{1}{2}\right]$$

Unit 1 AP Topics 14

4. Let f be the function given by $f(x)$=ln|x /1+x^2|

Find the range of f

Part D

2 points are earned for the answer

1 point deducted for error

max is

$$f(1) = \ln\frac{1}{2} = -\ln 2$$
$$\lim_{x\to 0^+} f(x) = -\infty$$
$$(or \lim_{x\to 0^-} f(x) = \lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = -\infty$$
$$\therefore range = \left(-\infty, \ln\frac{1}{2}\right]$$

0	1	2

The student response earns two of the following points:

2 points are earned for the answer

1 point deducted for error

max is

$$f(1) = \ln\frac{1}{2} = -\ln 2$$
$$\lim_{x\to 0^+} f(x) = -\infty$$
$$(or \lim_{x\to 0^-} f(x) = \lim_{x\to\infty} f(x) = \lim_{x\to-\infty} f(x) = -\infty$$
$$\therefore range = \left(-\infty, \ln\frac{1}{2}\right]$$

Unit 1 AP Topics 14

5. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \frac{2x^2+5x-3}{x^2+4x+3} & \text{for } x < -3 \\ kx + \frac{1}{2} & \text{for } -3 \le x \le 0 \\ \frac{2^x}{3^x-1} & \text{for } x > 0 \end{cases}$$

Let f be the function defined above, where k is a constant.

(a) For what value of k, if any, is f continuous at $x = -3$? Justify your answer.

(b) What type of discontinuity does f have at $x = 0$? Give a reason for your answer.

(c) Find all horizontal asymptotes to the graph of f. Show the work that leads to your answer.

Part A

At most 2 out of 3 points if mathematical notation for limits is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ $\lim\limits_{x\to-3^-} f(x)$
- ☐ $\lim\limits_{x\to-3^+} f(x)$
- ☐ justification

Solution:

$$\lim_{x\to-3^-} f(x) = \lim_{x\to-3^-} \frac{2x^2+5x-3}{x^2+4x+3} = \lim_{x\to-3^-} \frac{(2x-1)(x+3)}{(x+1)(x+3)}$$

$$= \lim_{x\to-3^-} \frac{2x-1}{x+1} = \frac{\lim\limits_{x\to-3^-}(2x-1)}{\lim\limits_{x\to-3^-}(x+1)}$$

$$= \frac{2(-3)-1}{-3+1} = \frac{7}{2}$$

$$\lim_{x\to-3^+} f(x) = \lim_{x\to-3^+} \left(kx + \frac{1}{2}\right) = -3k + \frac{1}{2}$$

$$f(-3) = -3k + \frac{1}{2}$$

f is continuous at $x = -3$ if $\lim\limits_{x\to-3^-} f(x) = \lim\limits_{x\to-3^+} f(x) = f(-3)$.

Unit 1 AP Topics 14

$-3k+\frac{1}{2}=\frac{7}{2} \Rightarrow -3k=3 \Rightarrow k=-1$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

he student response accurately includes both of the criteria below.

- [ ] $\lim\limits_{x\to 0^+} f(x)$
- [ ] answer

Solution:

$\lim\limits_{x\to 0^+} f(x) = \lim\limits_{x\to 0^+} \frac{2^x}{3^x-1} = \infty$ because the numerator approaches 1, while the denominator approaches 0 from the right.

Therefore, f has a discontinuity due to a vertical asymptote at $x=0$.

Part C

At most 1 out of 2 points if both correct asymptotes are listed without reference to limits OR if one correct asymptote is listed without any incorrect asymptotes and/or reference to limits.

Students do not have to show the factorization to earn the second point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $\lim\limits_{x\to\infty} f(x)$
- [ ] $\lim\limits_{x\to-\infty} f(x)$

Solution:

$\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty} \frac{2^x}{3^x-1} = \lim\limits_{x\to\infty} \frac{2^x}{3^x} = \lim\limits_{x\to\infty} \left(\frac{2}{3}\right)^x = 0$ because $0<\frac{2}{3}<1$.

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{2x^2+5x-3}{x^2+4x+3} = \lim_{x\to-\infty} \frac{x^2\left(2+\frac{5}{x}-\frac{3}{x^2}\right)}{x^2\left(1+\frac{4}{x}+\frac{3}{x^2}\right)}$$
$$= \lim_{x\to-\infty} \frac{2+\frac{5}{x}-\frac{3}{x^2}}{1+\frac{4}{x}+\frac{3}{x^2}} = \frac{\lim\limits_{x\to-\infty}\left(2+\frac{5}{x}-\frac{3}{x^2}\right)}{\lim\limits_{x\to-\infty}\left(1+\frac{4}{x}+\frac{3}{x^2}\right)}$$
$$= \frac{2+0-0}{1+0+0} = 2$$

Unit 1 AP Topics 14

The graph of f has horizontal asymptotes at $y = 0$ and $y = 2$.

Unit 1 AP Topics 14

6. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be the function defined by $f(x) = \frac{cx-5x^2}{2x^2+ax+b}$, where a, b, and c are constants. The graph of f has a vertical asymptote at $x = 1$, and f has a removable discontinuity at $x = -2$.

(a) Show that $a = 2$ and $b = -4$.

(b) Find the value of c. Justify your answer.

(c) To make f continuous at $x = -2$, $f(-2)$ should be defined as what value? Justify your answer.

(d) Write an equation for the horizontal asymptote to the graph of f. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ sets $2x^2 + ax + b = 0$ for $x = -2$ and $x = 1$
- ☐ verification

Solution:

For f to have a removable discontinuity at $x = -2$ the denominator must equal zero at $x = -2$.

$2(-2)^2 + a(-2) + b = 0 \Rightarrow 2a - b = 8$

For f to have a vertical asymptote at $x = 1$ the denominator must equal zero at $x = 1$.

$2(1)^2 + a(1) + b = 0 \Rightarrow a + b = -2$

$2a - b = 8$ and $a + b = -2 \Rightarrow 3a = 6 \Rightarrow a = 2$

$4 - b = 8 \Rightarrow b = 4 - 8 = -4$

Part B

Work for the justification point may come before the answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 14

0	1	2

The student response accurately includes both of the criteria below.

- [ ] answer
- [ ] justification

Solution:

Since f has a removable discontinuity at $x = -2$ and the denominator equals 0 at $x = -2$, the numerator must also equal 0 at $x = -2$.

$$0 = c(-2) - 5(-2)^2 = -2c - 20$$
$$\Rightarrow \ c = \frac{20}{-2} = -10$$

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] uses $\lim\limits_{x \to -2} f(x)$
- [ ] answer with justification

Solution:

f will be continuous at $x = -2$. if $f(-2) = \lim\limits_{x \to -2} f(x)$.

$$\lim_{x \to -2} f(x) = \lim_{x \to -2} \frac{-10x - 5x^2}{2x^2 + 2x - 4} = \lim_{x \to -2} \frac{-5x(2+x)}{2(x-1)(x+2)}$$
$$= \lim_{x \to -2} \frac{-5x}{2(x-1)} = \frac{-5(-2)}{2(-2-1)} = \frac{10}{-6} = -\frac{5}{3}$$

$f(-2)$ should be defined as $-\frac{5}{3}$ to make f continuous at $x = -2$.

Part D

No point if correct asymptote is listed without reference to limits.

Students do not have to show the factorization to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Unit 1 AP Topics 14

0	1

The student response accurately includes an equation for the horizontal asymptote.

Solution:

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{-10x-5x^2}{2x^2+2x-4} = \lim_{x\to\infty} \frac{x^2\left(-\frac{10}{x}-5\right)}{x^2\left(2+\frac{2}{x}-\frac{4}{x^2}\right)}$$

$$= \lim_{x\to\infty} \frac{-\frac{10}{x}-5}{2+\frac{2}{x}-\frac{4}{x^2}} = \frac{\lim_{x\to\infty}\left(-\frac{10}{x}-5\right)}{\lim_{x\to\infty}\left(2+\frac{2}{x}-\frac{4}{x^2}\right)}$$

$$= \frac{0-5}{2+0-0} = -\frac{5}{2}$$

The horizontal asymptote to the graph of f is $y = -\frac{5}{2}$.

Unit 1 AP Topics 14

7. Let *f* be the function given by $f(x) = \frac{\ln x}{x}$ mfrac for all $x > 0$. The derivative of *f* is given by $f'(x) = \frac{1-\ln x}{x^2}$.

(a) *Write an equation for the line tangent to the graph of f at x = e* ^{2}.

(b) Find the x-coordinate of the critical point of *f*. Determine whether this point is a relative minimum, a relative maximum, or neither for the function *f*. Justify your answer.

(c) The graph of the function *f* has exactly one point of inflection. Find the x-coordinate of this point.

(d) Find $\lim\limits_{x \to 0^+} f(x)$.

Part A

1 point is earned for the $f\left(e^2\right)$ and $f'\left(e^2\right)$

1 point is earned for the answer

$$f\left(e^2\right) = \frac{\ln e^2}{e^2} = \frac{2}{e^2},\ f'\left(e^2\right) = \frac{1-\ln e^2}{\left(e^2\right)^2} = -\frac{1}{e^4}$$

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}\left(x - e^2\right)$.

✓

0	1	2

The student response earns all of the following points:

1 point is earned for the $f\left(e^2\right)$ and $f'\left(e^2\right)$

1 point is earned for the answer

$$f\left(e^2\right) = \frac{\ln e^2}{e^2} = \frac{2}{e^2},\ f'\left(e^2\right) = \frac{1-\ln e^2}{\left(e^2\right)^2} = -\frac{1}{e^4}$$

An equation for the tangent line is $y = \frac{2}{e^2} - \frac{1}{e^4}\left(x - e^2\right)$.

Part B

3 points can be earned.

1 point for $x = e$

1 point for relative maximum

1 point for justification

Unit 1 AP Topics 14

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.

0	1	2	3

The student response earns all of the following points:

3 points can be earned.

1 point for $x = e$

1 point for relative maximum

1 point for justification

$f'(x) = 0$ when $x = e$. The function f has a relative maximum at $x = e$ because $f'(x)$ changes from positive to negative at $x = e$.

Part C

2 points are earned for $f''(x)$

1 point is earned for answer

$$f''(x) = \frac{-\frac{1}{x}x^2-(1-\ln x)2x}{x^4} = \frac{-3+2\ln x}{x^3} \text{ for all } x > 0$$
$$f''(x) = 0 \text{ when } -3+2\ln x = 0$$
$$x = e^{3/2}$$

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

0	1	2	3

The student response earns all of the following points:

2 points are earned for $f''(x)$

1 point is earned for answer

$$f''(x) = \frac{-\frac{1}{x}x^2-(1-\ln x)2x}{x^4} = \frac{-3+2\ln x}{x^3} \text{ for all } x > 0$$
$$f''(x) = 0 \text{ when } -3+2\ln x = 0$$
$$x = e^{3/2}$$

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$.

Part D

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1 point is earned for answer

$\lim_{x \to 0^+} \frac{\ln x}{x} = -\infty$ or Does Not Exist

0	1

The student response earns all of the following points:

1 point is earned for answer

$\lim_{x \to 0^+} \frac{\ln x}{x} = -\infty$ or Does Not Exist

Unit 1 AP Topics 14

8. Let f be the function that is given by $f(x) = ax + b / x^2 - c$ and that has the following properties.

(i) The graph of f is symmetric with respect to the y-axis.

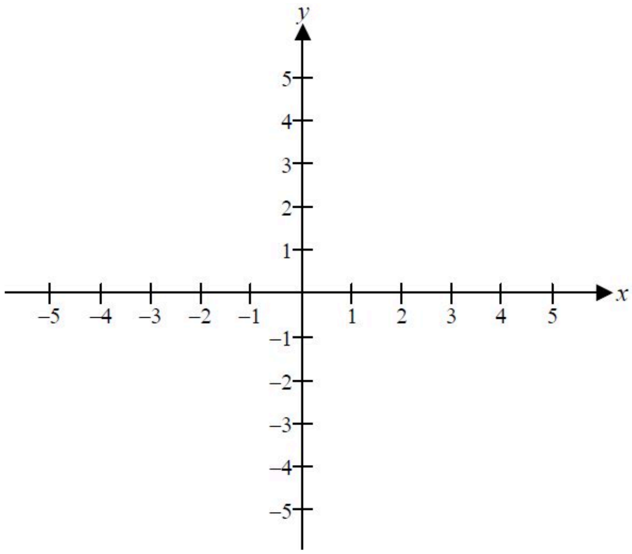
(ii) $\lim_{x \to 2^+} f(x) = +\infty$

(iii) $f'(1) = -2$

(a) Determine the values of a, b, and c.

(b) Write an equation for each vertical and each horizontal asymptote of the graph of f.

(c) Sketch the graph of f in the xy-plane provided below.



Part A

1 point is earned for $a = 0$

1 point is earned for $c = 4$

1 point is earned for $f'(x)$

1 point is earned for finding $b = 5$

If graph symmetric to y-axis then f is even

$f(-x) = f(x) \quad \therefore \boxed{a = 0}$

$\lim_{x \to 2^+} f(x) = +\infty \quad \therefore \boxed{c = 4}$

$f(x) = \frac{b}{x^2 - 4}$

$f'(x) = \frac{-2bx}{(x^2-4)^2}$

$-2 = f'(1) = \frac{-2b}{9} \quad \therefore \boxed{b = 9}$

0	1	2	3	4

The student response earns all of the following points:

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1 point is earned for $a = 0$

1 point is earned for $c = 4$

1 point is earned for $f'(x)$

1 point is earned for finding $b = 5$

If graph symmetric to y-axis then f is even

$$f(-x) = f(x) \qquad \therefore \boxed{a = 0}$$
$$\lim_{x \to 2^+} f(x) = +\infty \qquad \therefore \boxed{c = 4}$$
$$f(x) = \frac{b}{x^2 - 4}$$
$$f'(x) = \frac{-2bx}{(x^2-4)^2}$$
$$-2 = f'(1) = \frac{-2b}{9} \qquad \therefore \boxed{b = 9}$$

Part B

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

$$f(x) = \frac{9}{x^2 - 4}$$

VERTICAL: $x = 2$, $x = -2$

HORIZONTAL: $y = 0$

0	1	2

The student response earns all of the following points:

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

$$f(x) = \frac{9}{x^2 - 4}$$

VERTICAL: $x = 2$, $x = -2$

HORIZONTAL: $y = 0$

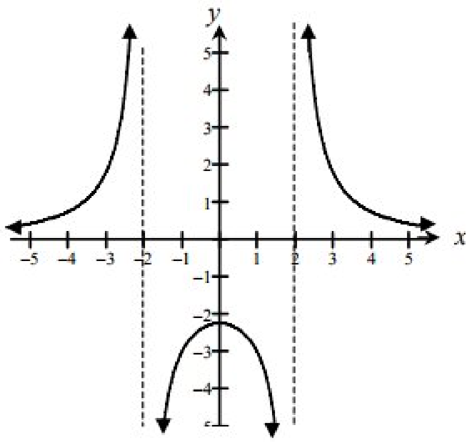
Part C

1 point is earned if symmetry agrees with their (a)

Unit 1 AP Topics 14

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y-int



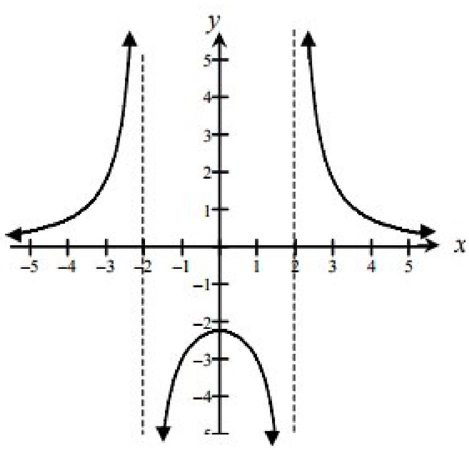
0	1	2	3

The student response earns all of the following points:

1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y-int



Unit 1 AP Topics 14

Let f be the function that is given by f(x) = $ax + b$ / $x^2 - c$ and that has the following properties.

(i) The graph of f is symmetric with respect to the y-axis.

(ii) $\lim_{x \to 2^+} f(x) = +\infty$

(iii) f'(1)=−2

9. Write an equation for each vertical and each horizontal asymptote of the graph of f.

Part B

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if aymptotes are switched

$f(x) = \frac{9}{x^2 - 4}$
Vertical : $x = 2, x = -2$
Horizontal : $y = 0$

0	1	2

The student response earns two of the following points:

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if aymptotes are switched

$f(x) = \frac{9}{x^2 - 4}$
Vertical : $x = 2, x = -2$
Horizontal : $y = 0$

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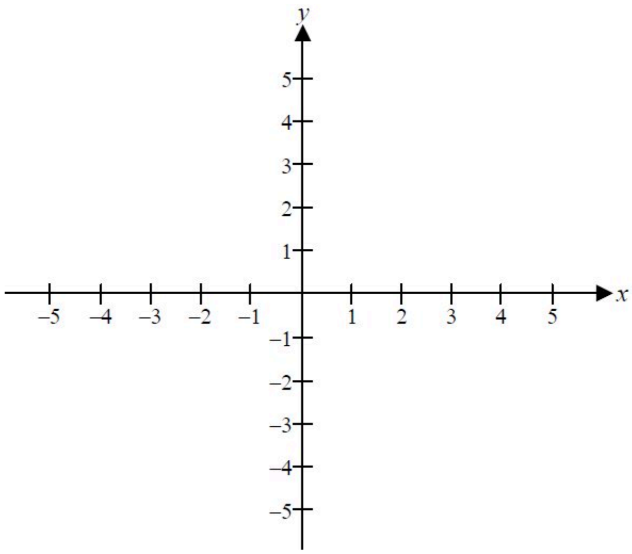
10. Sketch the graph of *f* in the *xy*-plane provided below.

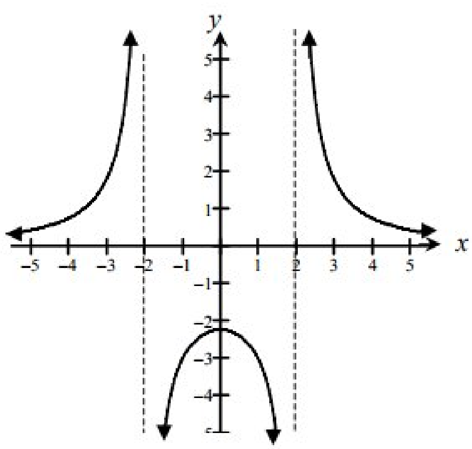


Part C

1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y-int



0	1	2	3

The student response earns three of the following points:

1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y-int

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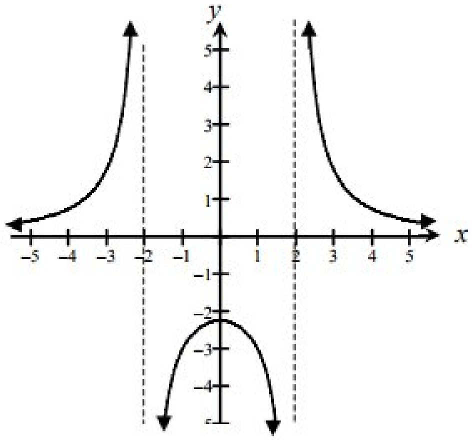
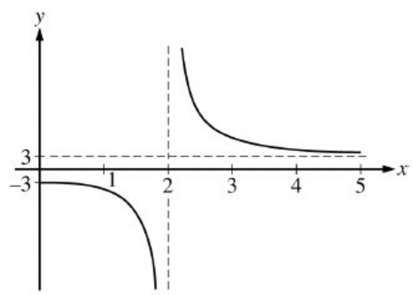


11.



The function f is given by $f(x) = \frac{ax^2+12}{x^2+b}$. The figure above shows a portion of the graph of f. Which of the following could be the values of the constants a and b?

(A) a = –3, b = 2

(B) a = 2, b = –3

(C) a = 2, b = –2

(D) a = 3, b = –4

(E) a = 3, b = 4

12. The continuous function f is positive and has domain $x > 0$. If the asymptotes of the graph of f are $x = 0$ and $y = 2$, which of the following statements must be true?

(A) $\lim_{x\to 0^+} f(x) = \infty$ and $\lim_{x\to 2} f(x) = \infty$

(B) $\lim_{x\to 0^+} f(x) = 2$ and $\lim_{x\to \infty} f(x) = 0$

(C) $\lim_{x\to 0^+} f(x) = \infty$ and $\lim_{x\to \infty} f(x) = 2$

(D) $\lim_{x\to 2} f(x) = \infty$ and $\lim_{x\to \infty} f(x) = 2$

13. The continuous function f is positive and has domain $x > 0$. If the asymptotes of the graph of f are $x = 0$ and $y = 2$, which of the following statements must be true?

(A) $\lim_{x\to 0^+} f(x) = \infty$ and $\lim_{x\to 2} f(x) = \infty$

(B) $\lim_{x\to 0^+} f(x) = 2$ and $\lim_{x\to \infty} f(x) = 0$

(C) $\lim_{x\to 0^+} f(x) = \infty$ and $\lim_{x\to \infty} f(x) = 2$

(D) $\lim_{x\to 2} f(x) = \infty$ and $\lim_{x\to \infty} f(x) = 2$

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14. Let *f* be the function given by $f(x) = \frac{x}{\sqrt{x^2-4}}$.

(a) Find the domain of *f*.

(b) Write an equation for each vertical asymptote to the graph of *f*.

(c) Write an equation for each horizontal asymptote to the graph of *f*.

(d) Find *f*′(*x*).

Part A

1 point is earned for $x^2 - 4 > 0$

1 point is earned for correct solution

$\boxed{x < -2 \text{ or } x > 2}$
OR $|x| > 2$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for $x^2 - 4 > 0$

1 point is earned for correct solution

$\boxed{x < -2 \text{ or } x > 2}$
OR $|x| > 2$

Part B

1 point is earned for finding value of *x*

$\boxed{x = 2, x = -2}$

0	1 ✓

The student response earns all of the following points:

1 point is earned for finding value of *x*

$\boxed{x = 2, x = -2}$

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Part C

1 point is earned for analysis in both direction

1 point is earned for 1st consistent solution

1 point is earned for 2nd consistent solution

$$\lim_{x\to+\infty}\frac{x}{\sqrt{x^2-4}}=1$$
$$\lim_{x\to-\infty}\frac{1}{\sqrt{x^2-4}}=-1$$
$$\boxed{y=1, y=-1}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for analysis in both direction

1 point is earned for 1st consistent solution

1 point is earned for 2nd consistent solution

$$\lim_{x\to+\infty}\frac{x}{\sqrt{x^2-4}}=1$$
$$\lim_{x\to-\infty}\frac{1}{\sqrt{x^2-4}}=-1$$
$$\boxed{y=1, y=-1}$$

Part D

3 points are earned for finding $f'(x)$

2 points are **deducted** for Chain rule error, Quotient (Product) rule error; and calculus error in exponent

1 point is **deducted** for all other errors

$$\begin{aligned} f'(x) &= \frac{\sqrt{x^2-4}-x\left[\frac{1}{2}\left(x^2-4\right)^{-1/2}2x\right]}{x^2-4} \\ &= \frac{\sqrt{x^2-4}-\frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\ &= \frac{-4}{\left(x^2-4\right)^{3/2}} \end{aligned}$$

0	1	2	3

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The student response earns all of the following points:

3 points are earned for finding $f'(x)$

2 points are **deducted** for Chain rule error, Quotient (Product) rule error; and calculus error in exponent

1 point is **deducted** for all other errors

$$\begin{aligned} f'(x) &= \frac{\sqrt{x^2-4} - x\left[\frac{1}{2}\left(x^2-4\right)^{-1/2} 2x\right]}{x^2-4} \\ &= \frac{\sqrt{x^2-4} - \frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\ &= \frac{-4}{\left(x^2-4\right)^{3/2}} \end{aligned}$$

15. The function g is defined by $g(x) = \frac{|x-5|}{x-5} \ln\left(\frac{x+5}{x^2}\right)$. At what values of x does the graph of g have a vertical asymptote?

(A) $x = -5$ only

(B) $x = 0$ only

(C) $x = -5$ and $x = 0$ only ✓

(D) $x = -5$, $x = 0$, and $x = 5$

Answer C

Correct. As x approaches -5 from the right, $\frac{x+5}{x^2}$ approaches 0 from the right and $\frac{|x-5|}{x-5} = -1$. Therefore $\lim\limits_{x \to -5^+} g(x) = -\lim\limits_{x \to 0^+} (\ln x) = +\infty$, which means that the graph of g has a vertical asymptote at $x = -5$.

As x approaches 0 from both the left and the right, $\frac{x+5}{x^2}$ approaches $+\infty$ and $\frac{|x-5|}{x-5} = -1$. Therefore $\lim\limits_{x \to 0} g(x) = -\lim\limits_{x \to \infty} (\ln x) = -\infty$, which means that the graph of g has a vertical asymptote at $x = 0$.

The graph of g has a jump discontinuity at $x = 5$, not a vertical asymptote, since $\lim\limits_{x \to 5^-} g(x) = -\ln\left(\frac{2}{5}\right)$ and $\lim\limits_{x \to 5^+} g(x) = \ln\left(\frac{2}{5}\right)$.

16. The function g is continuous at all x except $x = 4$. If $\lim\limits_{x \to 4} g(x) = \infty$, which of the following statements about g must be true?

(A) $g(4) = \infty$

(B) The line $x = 4$ is a horizontal asymptote to the graph of g.

(C) The line $x = 4$ is a vertical asymptote to the graph of g. ✓

(D) The line $y = 4$ is a vertical asymptote to the graph of g.

Answer C

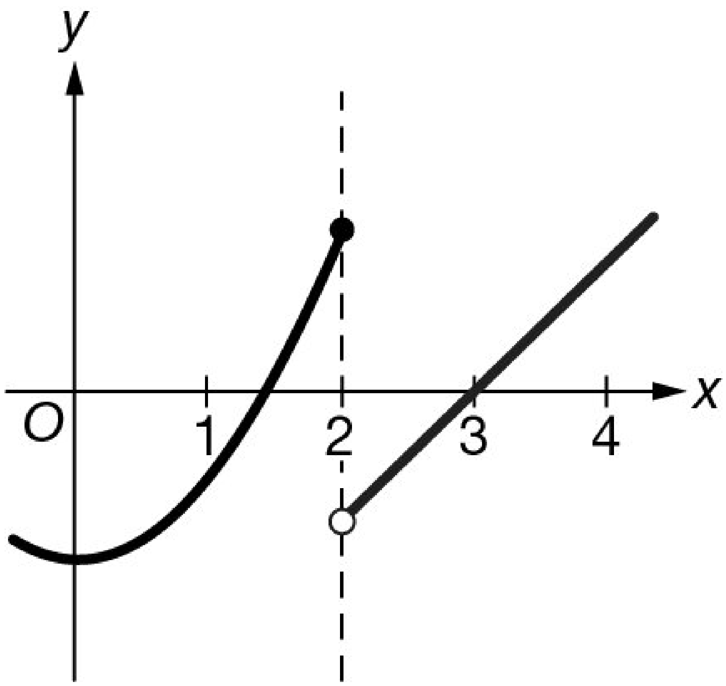
Correct. The unbounded behavior of $g(x)$ as x approaches 4 describes a vertical asymptote at $x = 4$.

Unit 1 AP Topics 14

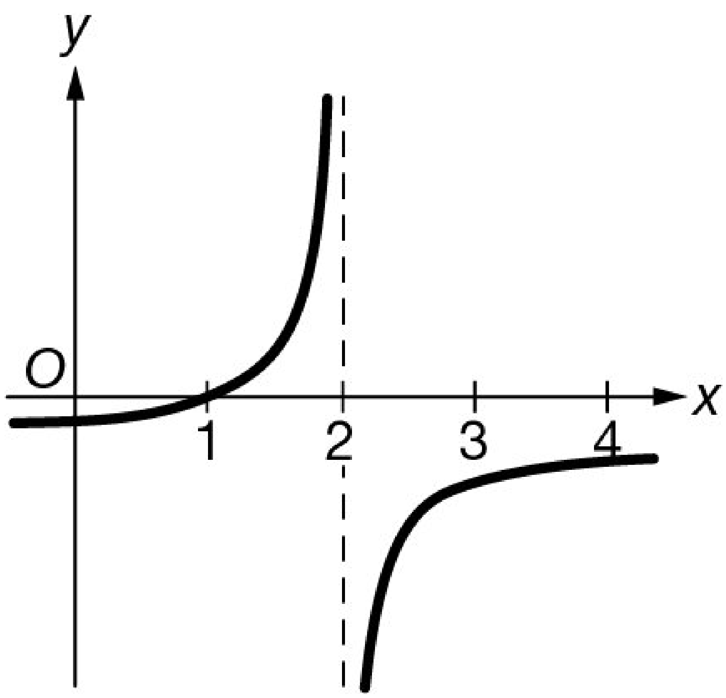
17. Let f be a function of x. If $\lim_{x\to 2^-} f(x) = +\infty$ and $\lim_{x\to 2^+} f(x) = +\infty$, which of the following could be a graph of f ?

Unit 1 AP Topics 14

(A)

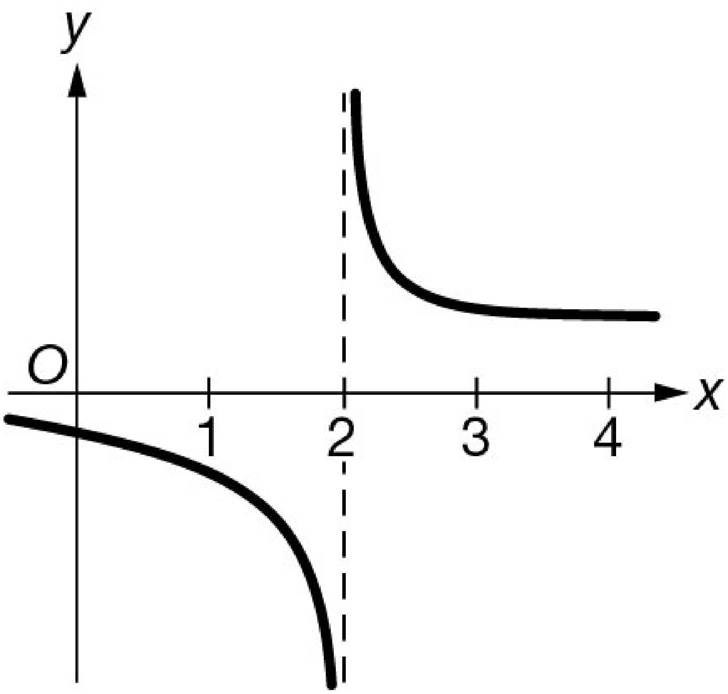


(B)

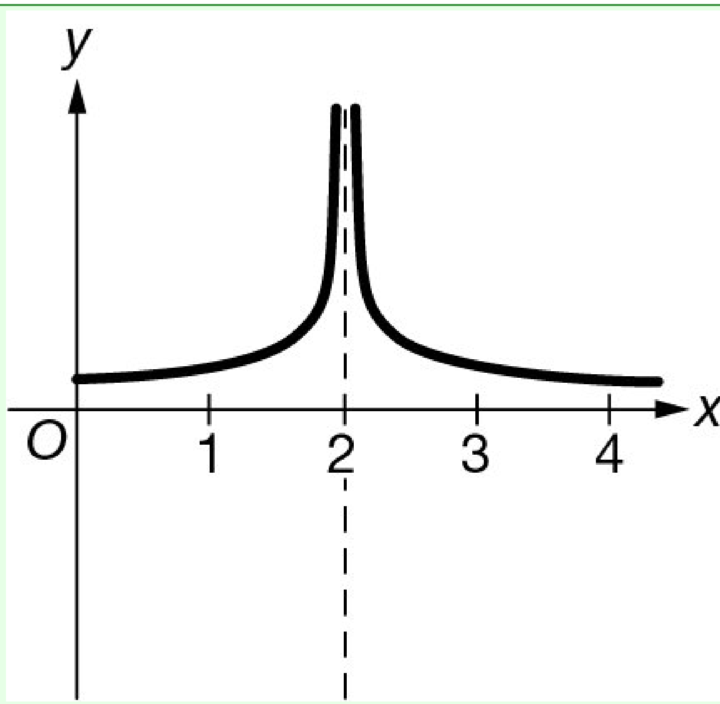


Unit 1 AP Topics 14

(C)



(D)



Answer D

Correct. The directions in which this graph approaches the vertical asymptote are $\lim_{x \to 2^-} f(x) = +\infty$ and $\lim_{x \to 2^+} f(x) = +\infty$.

Unit 1 AP Topics 14

18. Let f be a function such that $\lim_{x \to 3^-} f(x) = \infty$. Which of the following statements must be true?

(A) $\lim_{x \to 3^+} f(x) = \infty$

(B) f is undefined at $x = 3$.

(C) The graph of f has a vertical asymptote at $x = 3$.

(D) The graph of f has a vertical asymptote at $x = -3$.

Answer C

Correct. This statement is true. Because $f(x)$ increases without bound as x approaches 3 from the left, the graph of f must have a vertical asymptote at $x = 3$.

19. The function h is defined by $h(x) = \frac{x^2-9}{x-4}$. Which of the following statements must be true?

(A) $\lim_{x \to 4^-} h(x) = -\infty$ and $\lim_{x \to 4^+} h(x) = -\infty$

(B) $\lim_{x \to 4^-} h(x) = +\infty$ and $\lim_{x \to 4^+} h(x) = -\infty$

(C) $\lim_{x \to 4^-} h(x) = -\infty$ and $\lim_{x \to 4^+} h(x) = +\infty$

(D) $\lim_{x \to 4^-} h(x) = +\infty$ and $\lim_{x \to 4^+} h(x) = +\infty$

Answer C

Correct. As x approaches 4 from the left, $x^2 - 9$ is positive and approaches 7, while $x - 4$ is negative and approaches 0. Therefore, h is negative and approaches $-\infty$. As x approaches 4 from the right, $x^2 - 9$ is positive and approaches 7, while $x - 4$ is positive and approaches 0. Therefore, h is positive and approaches $+\infty$.

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20. The function g is defined by $g(x) = \left(\frac{x^2-3x-10}{x-5}\right) \ln\left(\frac{x^2+6x+9}{x^3+3x^2}\right)$. At what values of x does the graph of g have a vertical asymptote?

(A) $x = -3$ only

(B) $x = 0$ only

(C) $x = -3$ and $x = 0$ only ✓

(D) $x = -3$, $x = 0$, and $x = 5$

Answer C

Correct. As x approaches -3 from the right,
$\frac{x^2+6x+9}{x^3+3x^2} = \frac{(x+3)^2}{x^2(x+3)} = \frac{x+3}{x^2}$
approaches 0 from the right and at $x = -3$,
$\frac{x^2-3x-10}{x-5} = \frac{9+9-10}{-3-5} = \frac{8}{-8} = -1$.
Therefore, $\lim\limits_{x\to -3^+} g(x) = -\lim\limits_{x\to 0^+} (\ln x) = \infty$. As x approaches 0 from both the left and the right,
$\frac{x^2+6x+9}{x^3+3x^2} = \frac{(x+3)^2}{x^2(x+3)} = \frac{(x+3)}{x^2}$
approaches $+\infty$ and at $x = 0$, $\frac{x^2-3x-10}{x-5} = \frac{-10}{-5} = 2$. Therefore, $\lim\limits_{x\to 0} g(x) = -\lim\limits_{x\to\infty} (\ln x) = +\infty$. The graph of g has a removable discontinuity at $x = 5$, not a vertical asymptote, since $g(5)$ is undefined but
$\lim\limits_{x\to 5} \frac{x^2-3x-10}{x-5} = \lim\limits_{x\to 5} \frac{(x+2)(x-5)}{x-5} = \lim\limits_{x\to 5}(x+2) = 7$
and $\ln\left(\frac{5^2+6(5)+9}{5^3+3(75)}\right)$ is finite.

21. The function g is continuous at all x except $x = 2$. If $\lim\limits_{x\to 2} g(x) = \infty$, which of the following statements about g must be true?

(A) $g(2) = \infty$

(B) The line $x = 2$ is a horizontal asymptote to the graph of g.

(C) The line $x = 2$ is a vertical asymptote to the graph of g. ✓

(D) The line $y = 2$ is a vertical asymptote to the graph of g.

Answer C

Correct. The unbounded behavior of $g(x)$ as x approaches 2 describes a vertical asymptote at $x = 2$.

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22. Let f be a function such that $\lim_{x \to 5^-} f(x) = \infty$. Which of the following statements must be true?

(A) $\lim_{x \to 5^+} f(x) = \infty$

(B) f is undefined at $x = 5$.

(C) The graph of f has a vertical asymptote at $x = 5$.

(D) The graph of f has a vertical asymptote at $x = -5$.

Answer C

Correct. This statement is true. Because $f(x)$ increases without bound as x approaches 5 from the left, the graph of f must have a vertical asymptote at $x = 5$.

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23. Let f be a function of x. If $\lim_{x \to 2^-} f(x) = +\infty$ and $\lim_{x \to 2^+} f(x) = -\infty$, which of the following could be a graph of f ?

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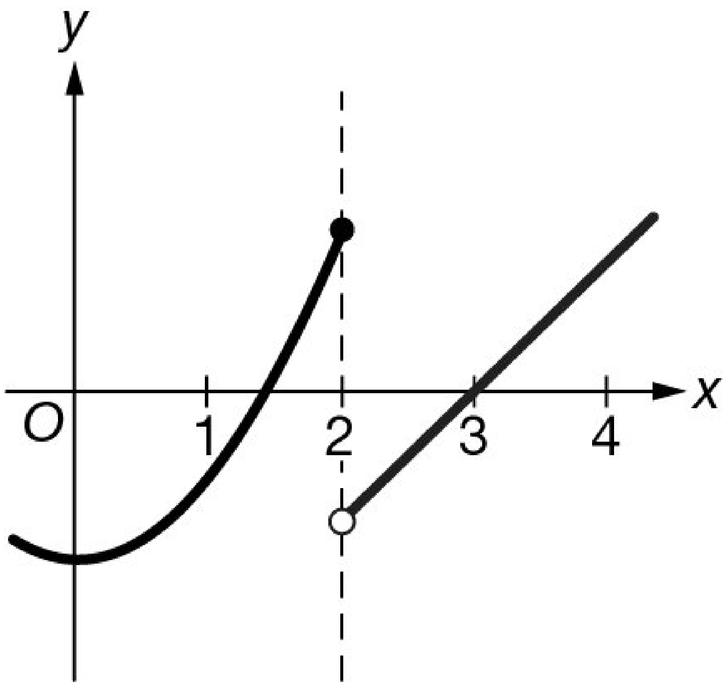
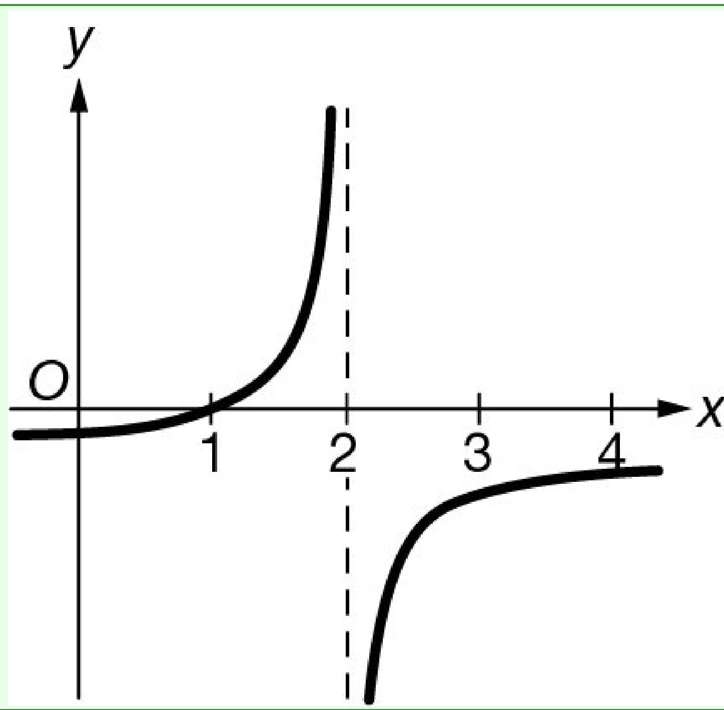
(A)



(B)



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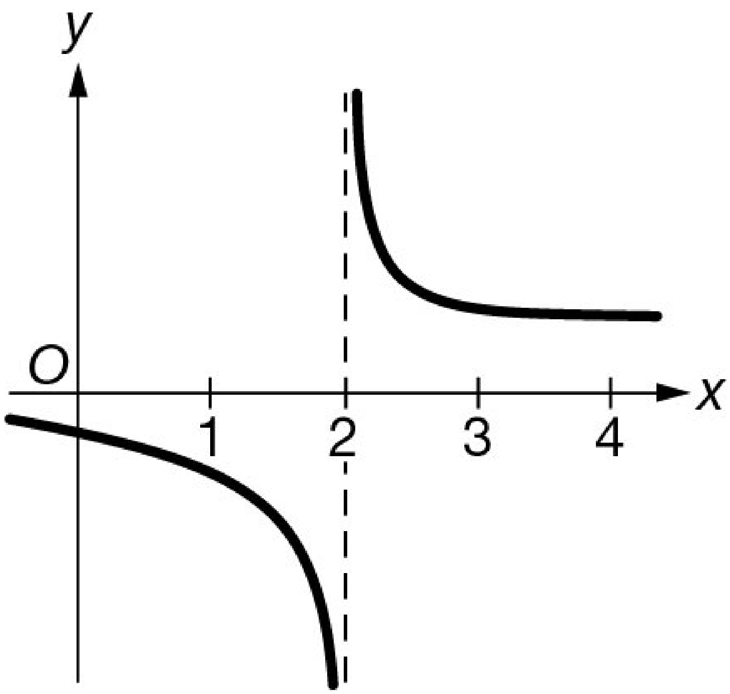
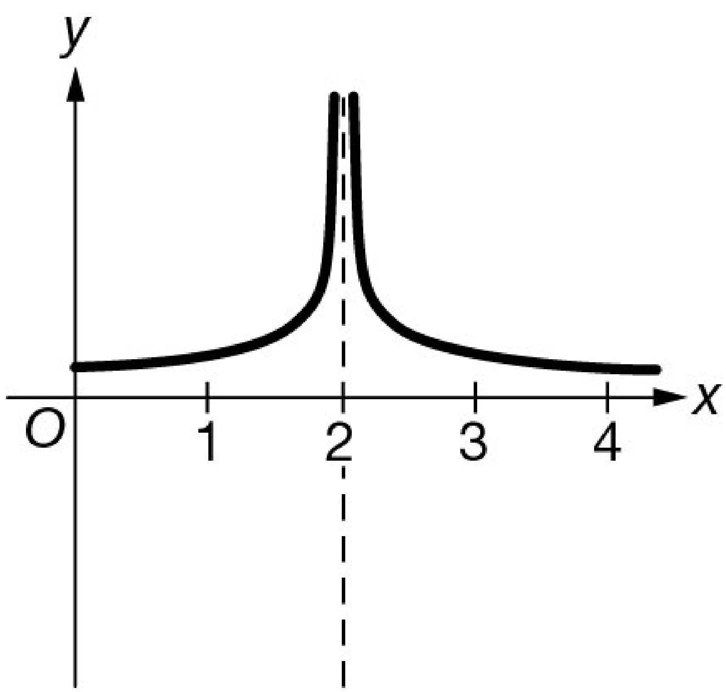
(C)



(D)



Answer B

Correct. The graph of f has a vertical asymptote at $x = 2$, and the graph of f should approach the asymptote in opposite directions. The directions in which this graph approaches the vertical asymptote matches the conditions that $\lim_{x \to 2^-} f(x) = +\infty$ and $\lim_{x \to 2^+} f(x) = -\infty$.

Unit 1 AP Topics 14

24. The values $f(x)$ of a function f can be made arbitrarily large by taking x sufficiently close to 1 but not equal to 1. Which of the following statements must be true?

(A) $f(1)$ does not exist.

(B) f is continuous at $x = 1$.

(C) $\lim_{x \to 1} f(x) = \infty$

(D) $\lim_{x \to \infty} f(x) = 1$

Answer C

Correct. The concept of a limit can be extended to include infinite limits. Since $f(x)$ can be made arbitrarily large as x approaches 1, the values of f become unbounded as x approaches 1. This means that the limit of $f(x)$ is infinite as x approaches 1.

25. The values $f(x)$ of a function f can be made arbitrarily large by taking x sufficiently close to 2 but not equal to 2. Which of the following statements must be true?

(A) $f(2)$ does not exist.

(B) f is continuous at $x = 2$.

(C) $\lim_{x \to 2} f(x) = \infty$

(D) $\lim_{x \to \infty} f(x) = 2$

Answer C

Correct. The concept of a limit can be extended to include infinite limits. Since $f(x)$ can be made arbitrarily large as x approaches 2, the values of f become unbounded as x approaches 2. This means that the limit of $f(x)$ is infinite as x approaches 2.

26. Let *f* be the function given by $f(x) = \frac{x}{\sqrt{x^2-4}}$.

Write an equation for each vertical asymptote to the graph of *f*.

Part B

1 point is earned for the asymptotes

$x = 2, x = -2$

0	1

The student response earns one of the following points:

1 point is earned for the asymptotes

$x = 2, x = -2$

Unit 1 AP Topics 14

27. If the graph of $y=\frac{ax+b}{x+c}$ has a horizontal asymptote y=2 and a vertical asymptote x=-3, then $a+c=$

(A) -5

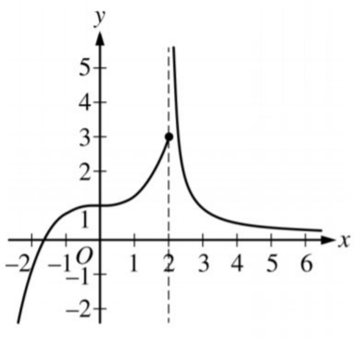
(B) -1

(C) 0

(D) 1

(E) 5 ✓

28.



Graph of f

The graph of the function f is shown in the figure above.

Which of the following statements must be false?

(A) $\lim_{x\to 2^-} f(x)=3$

(B) $\lim_{x\to 2^+} f(x)=\infty$

(C) $\lim_{x\to 2} f(x)=f(2)$ ✓

(D) $\lim_{x\to\infty} f(x)=0$

29. The vertical line $x = 2$ is an asymptote for the graph of the function f. Which of the following statements must be false?

(A) $\lim_{x\to 2} f(x)=0$ ✓

(B) $\lim_{x\to 2} f(x)=-\infty$

(C) $\lim_{x\to 2} f(x)=\infty$

(D) $\lim_{x\to\infty} f(x)=2$

(E) $\lim_{x\to\infty} f(x)=\infty$

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30. The function h is defined by $h(x) = \frac{x^2-7}{x-3}$. Which of the following statements must be true?

(A) $\lim_{x \to 3^-} h(x) = -\infty$ and $\lim_{x \to 3^+} h(x) = -\infty$

(B) $\lim_{x \to 3^-} h(x) = +\infty$ and $\lim_{x \to 3^+} h(x) = -\infty$

(C) $\lim_{x \to 3^-} h(x) = -\infty$ and $\lim_{x \to 3^+} h(x) = +\infty$

(D) $\lim_{x \to 3^-} h(x) = +\infty$ and $\lim_{x \to 3^+} h(x) = +\infty$

Answer C

Correct. As x approaches 3 from the left, $x^2 - 7$ is positive and approaches 2, while $x - 3$ is negative and approaches 0. Therefore h is negative and approaches $-\infty$. As x approaches 3 from the right, $x^2 - 7$ is positive and approaches 2, while $x - 3$ is positive and approaches 0. Therefore h is positive and approaches $+\infty$.

31. Let f be a continuous function such that $\int_0^{17} f(x)\,dx = 8$, $\int_{17}^{20} f(x)\,dx = -3$, and $\int_{13}^{20} f(x)\,dx = 7$. What is the value of $\int_0^{13} f(x)\,dx$?

(A) -2

(B) 4

(C) 12

(D) 18

Answer A

Correct. Using the property of definite integrals over adjacent intervals, $\int_0^{20} f(x)dx = \int_0^{17} f(x)dx + \int_{17}^{20} f(x)dx = 8 + (-3) = 5$.
Another application of the same property gives
$\int_0^{20} f(x)dx = \int_0^{13} f(x)dx + \int_{13}^{20} f(x)dx \Rightarrow \int_0^{13} f(x)dx = \int_0^{20} f(x)dx - \int_{13}^{20} f(x)dx$. Therefore,
$\int_0^{13} f(x)dx = \int_0^{20} f(x)dx - \int_{13}^{20} f(x)dx = 5 - 7 = -2$.

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32. The function f has the property that $\lim_{x \to 1^-} f(x) = +\infty$, $\lim_{x \to 1^+} f(x) = -\infty$, $\lim_{x \to -\infty} f(x) = 2$, and $\lim_{x \to +\infty} f(x) = 2$.

Of the following, which could be the graph of f ?

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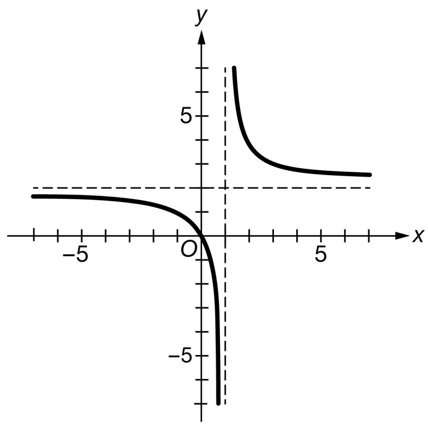
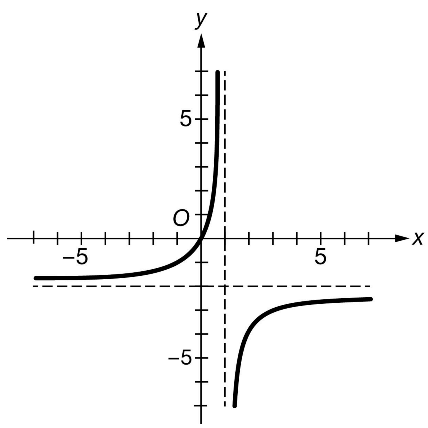
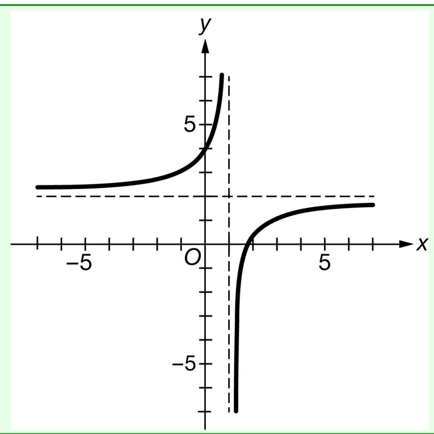
(A)



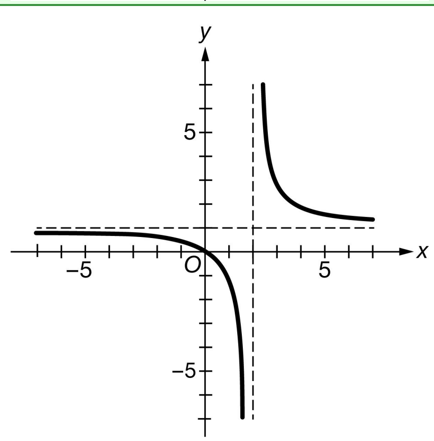
(B)



(C)



(D)



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Answer C

Correct. Since $\lim_{x \to 1^-} f(x) = +\infty$ and $\lim_{x \to 1^+} f(x) = -\infty$, the graph of f approaches the vertical asymptote at $x = 1$ in the upward direction as x approaches 1 from the left and approaches the vertical asymptote in the downward direction as x approaches 1 from the right. Since $\lim_{x \to -\infty} f(x) = 2$ and $\lim_{x \to +\infty} f(x) = 2$, the graph of f approaches the horizontal asymptote at $y = 2$ in both horizontal directions. This graph exhibits these properties and therefore could be the graph of f.

33. The table shown gives selected values for a rational function g.

x	-500	-100	-50	-10	0	10	50	100	500
$g(x)$	0.9996	0.9901	0.9615	0.5	0	0.5	0.9615	0.9901	0.9996

Of the following, which statement is supported by the data in the table?

(A) The graph of g has a vertical asymptote at $x = 0$.

(B) The graph of g has a vertical asymptote at $x = 500$.

(C) The graph of g has a horizontal asymptote at $y = 0$.

(D) The graph of g has a horizontal asymptote at $y = 1$. ✓

Answer D

Correct. The data supports the conclusion that $\lim_{x \to \infty} g(x) = 1$ and $\lim_{x \to -\infty} g(x) = 1$. This would mean that the graph of g has a horizontal asymptote at $y = 1$.